

SIMC yield physics: from beam charge to a Monte

This note explains what a SIMC yield means, why it can be compared with a data yield, and how Ngen, Ntried, Weight, normfac, and fWeight fit together. The main example is a positive-Ngen SIDIS coincidence run.

The central idea is:

Data count physical interactions produced by a measured luminosity. SIMC numerically integrates a cross-section model over the same experimental phase space and detector acceptance. They can therefore be expressed in the same unit: predicted counts (or counts per unit charge).

SIMC does not need to reproduce individual beam bunches, ionization, or ADC pulses to make this prediction. It samples the variables in the event-rate integral and transports the resulting particles through a model of the spectrometers.

1. Quantities called "yield"

Several different quantities are casually called a yield. Keep their names and units explicit.

1.1 Data quantities

For one run or run group:

- N_raw: events passing the analysis cuts before background subtraction or efficiency correction.
- N_bg: estimated events from random coincidences, target walls, charge-symmetric background, or other sources that are not part of the desired reaction.
- N_net = N_raw - N_bg: background-subtracted events.
- Q: accepted integrated beam charge, commonly in mC.
- epsilon_data: product of the applicable data efficiencies and live-time factors.
- N_corr = N_net / epsilon_data: the number corrected to the chosen ideal

data convention.

- $Y_{\text{data}}^Q = N_{\text{corr}} / Q$: charge-normalized yield, for example counts/mC.

If the target density during production was a fraction f_{rho} of the target density used by SIMC, there are two equivalent conventions:

1. use the effective target thickness $f_{\text{rho}} * t_{\text{nominal}}$ in SIMC, and do not density-correct the data; or
2. use the nominal target thickness in SIMC and divide the data by f_{rho} .

Do one, not both. In convention 2 a schematic corrected data yield is

$$Y_{\text{data}}^Q = (N_{\text{raw}} - N_{\text{bg}}) / (Q * LT * \epsilon_{\text{trigger}} * \epsilon_{\text{track}} * \epsilon_{\text{PID}} * \epsilon_{\text{detector}} * f_{\text{rho}}).$$

Not every experiment uses exactly this bookkeeping. For example, a target- wall contribution is normally subtracted rather than represented by an "efficiency." The important requirement is that data and SIMC describe the same idealized measurement.

1.2 Luminosity and absolute yield

For an incident electron flux and a target with areal number density n_t ,

$$\mathcal{L}_{\text{int}} = N_e n_t, \quad N_e = \frac{Q}{e}.$$

For a differential cross section in variables Φ , the predicted count is

$$N_{\text{pred}} = \mathcal{L}_{\text{int}} \int_R \frac{d\sigma}{d\Phi} A(\Phi) \epsilon(\Phi) d\Phi.$$

Here R is the generated region, A includes geometrical and reconstruction acceptance, and ϵ contains effects represented in the simulation. The result is a number of counts, not a cross section.

The local SIMC source constructs a target conversion factor from the target mass, abundance, angle, and thickness, then computes

$$\text{luminosity} = \text{EXPER\%charge} / \text{targetfac}$$

in simc.f. EXPER%charge is in mC and luminosity is reported in microbarn inverse. Thus, charge is an actual SIMC input. Setting it to 1 mC asks SIMC for a prediction normalized to 1 mC; it does not mean that SIMC has simulated one physical coulomb, one beam bunch, or one electron at a time.

1.3 Two equivalent comparison conventions

One may compare absolute counts at the data charge:

$$N_{\text{data, corr}} \quad \text{with} \quad N_{\text{SIMC}}(Q_{\text{data}}),$$

or compare charge-normalized yields:

$$\frac{N_{\text{data, corr}}}{Q_{\text{data}}} \quad \text{with} \quad \frac{N_{\text{SIMC}}(Q_{\text{SIMC}})}{Q_{\text{SIMC}}}.$$

Because the SIMC luminosity is linear in EXPER%charge, these are equivalent when all other conventions match. Running SIMC with 1 mC simply makes its summed yield numerically equal to a prediction in counts/mC.

2. Why compare a SIMC yield with a data yield?

A cross section is not observed directly. The experiment observes events after finite acceptance, radiation, energy loss, resolution, reconstruction, and cuts. A point cross section evaluated only at the central kinematics does not generally predict that event count.

SIMC performs the convolution needed to connect the physics model to the measurement. Consequently, an absolute data/SIMC yield comparison tests the combined description of:

- integrated luminosity and target thickness;
- the differential cross-section model across the accepted phase space;
- radiative effects, energy loss, multiple scattering, and particle decay;
- spectrometer optics, apertures, reconstruction, and resolution;
- analysis cuts and the remaining data efficiency corrections.

This is both the strength and the limitation of a yield ratio. A discrepancy does not by itself identify which item is wrong. Kinematic distributions and systematic variations are needed for diagnosis.

Shape agreement alone is weaker than absolute agreement. Histograms can be arbitrarily area-normalized and have matching shapes even if the luminosity, cross-section magnitude, target thickness, or an overall efficiency is incorrect. Conversely, one integrated number can agree while compensating shape errors cancel across the acceptance. A useful validation checks both.

3. SIMC as Monte Carlo integration

3.1 The elementary estimator

Suppose SIMC throws points uniformly in a generated phase-space region with volume

$$V_{\text{gen}} = \int_R d\Phi.$$

Let l_i equal one when throw i survives the simulated acceptance and the analysis selection, and zero otherwise. Let w_i contain the physics integrand and any Jacobian or generation correction. Uniform Monte Carlo integration gives

$$N_{\text{pred}} \approx \mathcal{L}_{\text{int}} \frac{V_{\text{gen}}}{N_{\text{tried}}} \sum_{i=1}^{N_{\text{tried}}} l_i w_i = \mathcal{L}_{\text{int}} \frac{V_{\text{gen}}}{N_{\text{tried}}} \sum_{i \in \text{selected}} w_i.$$

This equation explains why failed throws matter even though no failed event appears in the final selected histogram: every random point belongs in the denominator N_{tried} . Replacing N_{tried} with the number of accepted events would remove the acceptance loss and bias the integral.

This also explains why an unweighted count of SIMC tree entries is not the SIMC yield. Tree entries are integration samples, not equivalent physical events.

3.2 Event lifecycle in the local SIMC implementation

For each loop iteration, `simc.f`:

1. increments `ntried`;
2. calls `generate`, which samples the generated variables and completes the reaction kinematics;
3. if generation succeeds, calls `montecarlo` to transport the particles through target and spectrometer effects;
4. reconstructs the event and checks spectrometer-population and physics

cuts;

5. writes a diagnostic ntuple entry according to the configured output mode;

6. increments the successful-event counter only according to the ngen

mode.

Within event generation and completion, the code can apply radiation, multiple scattering, energy loss, optics and apertures, decay, spectral or target factors, reaction cross sections, and coordinate Jacobians. SIMC is therefore an acceptance-aware physics Monte Carlo, but it is not a full GEANT detector-response simulation: ordinary data acquisition and detector efficiencies still need appropriate data-side treatment unless explicitly modeled.

4. Ntried, Ngen, successes, and tree entries

These quantities have different meanings.

4.1 Positive ngen

The input files document positive ngen as the requested number of successes. The event loop condition is

```
do while (nevent.lt.abs(ngen))
```

and, for positive ngen, nevent increments only when success is true. Therefore the run continues until

$$N_{\text{success}} = N_{\text{gen}},$$

while generally

$$N_{\text{tried}} \geq N_{\text{gen}}.$$

The ratio Ngen/Ntried is the overall success fraction for the stages that control success. It is not an efficiency to apply to data; it is part of the Monte Carlo integration bookkeeping.

4.2 Negative ngen

For negative ngen, nevent increments on every loop iteration, whether or not the throw succeeds. The program consequently makes exactly $\text{abs}(\text{ngen})$ attempts. This mode is useful when poor acceptance might otherwise make SIMC search indefinitely for a requested number of successes.

The recon_hcana helper only creates a nonzero fWeight when its parsed Ngen is positive. The positive-Ngen formula in this note must therefore not be blindly applied to a negative-Ngen output. For production yield work, use a positive ngen unless the downstream normalization has been explicitly designed for fixed-attempt mode.

4.3 ROOT entries and later analysis cuts

Ngen is a run-control count; Ntried is an attempted-throw count. A ROOT tree entry is an output record. Their equality depends on the output path and success/write logic. Furthermore, a later ROOT selection can retain only a subset of stored entries.

Never normalize by `tree->GetEntries()` merely because it is convenient. Use the SIMC normalization metadata, and calculate a selected-bin yield by summing the per-entry normalized weights for entries passing that bin's cuts.

5. SIDIS generation volume

The code approximates each arm's angular measure in target slopes as

$$\Delta\Omega_e = (y'_{e,\max} - y'_{e,\min})(x'_{e,\max} - x'_{e,\min}),$$

$$\Delta\Omega_h = (y'_{h,\max} - y'_{h,\min})(x'_{h,\max} - x'_{h,\min}).$$

For doing_semi, the local simc.f constructs

$$V_{\text{gen}} = \Delta\Omega_e \Delta\Omega_h \Delta E_{e'} \Delta E_h.$$

Thus the SIDIS generation is six-dimensional: two electron angles, scattered electron energy, two hadron angles, and hadron energy. In the code's native convention the angular slopes are in radians and the energy ranges are in MeV, so the generation volume carries the corresponding powers of sr and MeV. The units cancel those of the differential cross section and event weight so that the final luminosity-weighted result is a count.

The generated region is not identical to the accepted region. It should cover the relevant acceptance; transport and cuts then determine which throws survive.

For hydrogen $H(e,e'p)$, the generated volume starts with only the electron angular volume in this normalization block. Other reaction modes multiply different energy and hadron-angle factors, which is why one must not copy a SIDIS `genvol` formula into every reaction.

6. What the stored Weight means

Near the end of `complete_main` in `event.f`, the local code forms

```
main%weight = main%SF_weight*main%jacobian*main%gen_weight*main%sigcc
main%weight = main%weight * tgtweight
```

and, for a kaon mode in which decay is not explicitly simulated, also multiplies a survival probability. Schematically,

$$W_i = W_{SF,i} J_i W_{gen,i} \sigma_i W_{target,i} W_{survival,i}.$$

- `sigcc` is the reaction-dependent differential cross-section model at the event kinematics, including the code's configured corrections.
- `jacobian` transforms between the variables used by the cross-section model and those sampled or reconstructed by SIMC.
- `gen_weight` corrects event-by-event restrictions of a nominal generation interval. It starts at one and is modified when kinematics narrow an allowed energy interval.
- `SF_weight` represents the spectral-function factor for applicable nuclear reactions; it is one for modes such as SIDIS in this branch of the code.
- `tgtweight` converts an appropriate per-nucleon model to the configured target content where required.
- a survival factor can represent a decay probability when decay is weighted instead of explicitly generated.

Weight is therefore primarily the value of the integration integrand for one sampled point. It does not include the complete generated volume, the integrated luminosity, or the division by the total number of random throws. Consequently, `sum(Weight)` by itself is neither an absolute count nor a charge-normalized yield.

Acceptance is also not generally a continuous factor inside Weight. Transport failures and cuts most often act as a zero/one decision: a failed throw contributes zero, while a surviving throw contributes its weight.

7. normfac, fWeight, and the SIMC yield

For a standard positive-ngen run, simc.f calculates

```
normfac = luminosity/ntried*nevent
...
normfac = normfac * genvol
```

At normal completion nevent = Ngen, so

$$\text{normfac} = \mathcal{L}_{\text{int}} \frac{N_{\text{gen}}}{N_{\text{tried}}} V_{\text{gen}}.$$

Why put Ngen into normfac only to divide by it later? Historically, SIMC's integrated weight accumulator contains a sum over successful events, and this normalization converts its average-over-successes structure into the full attempted-volume integral. The ROOT helper exposes the same result as a convenient per-entry quantity:

```
fWeight = Weight * simc_normfactor / simc_nevents;
```

Algebraically,

$$\text{fWeight}_i = w_i \frac{\text{normfac}}{N_{\text{gen}}} = w_i \frac{\mathcal{L}_{\text{int}} V_{\text{gen}}}{N_{\text{tried}}}.$$

Therefore the predicted yield in any selected bin or region B is

$$Y_{\text{SIMC}, B} = \sum_{i \in B} \text{fWeight}_i.$$

One fWeight is one Monte Carlo sample's contribution to the integral. It is not the whole yield. The unweighted number of selected SIMC entries is also not the yield.

Important special mode: doing_phsp

The source explicitly sets $\text{normfac} = 1.0$ when doing_phsp is true. That mode is intended for phase-space studies and bypasses the standard absolute normalization. Do not interpret its $\text{Weight} \times \text{normfac} / N_{\text{gen}}$ as the production yield formula derived above without a separate definition for that study.

8. Worked SIDIS coincidence example

The following is a deliberately small toy example. Its units are chosen to be mutually consistent, but its numerical values are illustrative rather than representative of a particular experiment.

8.1 SIMC configuration and normalization

Suppose a SIDIS SIMC run uses:

```

EXPER%charge = 1.0 mC
luminosity   = 2500 /microbarn
Ntried       = 100000
Ngen         = 40000 successful events
genvol       = 0.020 sr^2 MeV^2

```

The success fraction is $40000/100000 = 0.4$, and

$$\text{normfac} = 2500 \times 0.4 \times 0.020 = 20.$$

The common per-event integration factor is

$$\frac{\text{normfac}}{N_{\text{gen}}} = \frac{20}{40000} = 5.0 \times 10^{-4},$$

which is also

$$\frac{\mathcal{L}V_{\text{gen}}}{N_{\text{tried}}} = \frac{2500 \times 0.020}{100000} = 5.0 \times 10^{-4}.$$

After a tight analysis-bin cut, suppose five stored events survive:

Event	Weight	fWeight = Weight x 0.0005 (counts at 1 mC)
1	520	0.260
2	480	0.240
3	650	0.325
4	450	0.225
5	700	0.350
Sum	2800	1.400

The SIMC result for the bin is therefore

$$Y_{\text{SIMC}}^Q = 1.400 \text{ counts/mC.}$$

This illustrates the role of rejected events. The final table contains only five selected entries, but the integration denominator remains 100000 total throws. Using five, 40000, or the total ROOT entries in its place would give a different and incorrect result.

8.2 Corrected data yield

Suppose the matching data sample has

```

Q                = 12.0 mC
N_raw            = 20 events
N_bg             = 2 events
DAQ live time    = 0.92
trigger efficiency = 0.99
tracking efficiency = 0.96
PID efficiency   = 0.98
detector efficiency = 0.995
relative density  = 0.97

```

Here SIMC uses the nominal target thickness, so the data are corrected to nominal density. The total correction factor is

$$\epsilon_{\text{data}} = 0.92 \times 0.99 \times 0.96 \times 0.98 \times 0.995 \times 0.97 = 0.827018.$$

Thus

$$N_{\text{data, corr}} = \frac{20 - 2}{0.827018} = 21.765$$

and

$$Y_{\text{data}}^Q = \frac{21.765}{12.0} = 1.814 \text{ counts/mC.}$$

8.3 Absolute-count and per-charge comparisons

Rescaling the 1 mC SIMC result to the data charge gives

$$N_{\text{SIMC}}(12 \text{ mC}) = 1.400 \times 12 = 16.800.$$

The absolute-count ratio is

$$R = \frac{21.765}{16.800} = 1.296.$$

The per-charge ratio is identically

$$R = \frac{1.814}{1.400} = 1.296.$$

The two conventions contain the same physics. A ratio above one means that, under the stated corrections and cuts, data exceed the SIMC prediction by about 30%. It does not prove that the cross-section model alone is low; luminosity, target, efficiencies, backgrounds, acceptance, and radiative modeling must also be considered.

8.4 Weighted Monte Carlo statistical uncertainty

For independent weighted samples, the usual Monte Carlo variance estimate for the selected sum is

$$\delta Y_{\text{SIMC}} = \sqrt{\sum_i f \text{Weight}_i^2}.$$

For the five toy events,

$$\delta Y_{\text{SIMC}} = \sqrt{0.260^2 + 0.240^2 + 0.325^2 + 0.225^2 + 0.350^2} = 0.636 \text{ counts/mC.}$$

The effective number of equally weighted events is

$$N_{\text{eff}} = \frac{(\sum_i f_i)^2}{\sum_i f_i^2} = \frac{1.400^2}{0.40395} = 4.85.$$

This intentionally tiny example has a 45% relative SIMC statistical error. A production run should generate enough events that this uncertainty is small compared with the desired experimental precision in every reported bin.

For a simplified illustration in which the background estimate has no uncertainty, take $\sqrt{N_{\text{net}}}$ as the data counting uncertainty. After the same corrections,

$$\delta Y_{\text{data}}^Q = \frac{\sqrt{18}}{12 \times 0.827018} = 0.428 \text{ counts/mC.}$$

If data and Monte Carlo statistics are independent,

$$\delta R_{\text{stat}} = R \sqrt{\left(\frac{\delta Y_{\text{data}}}{Y_{\text{data}}}\right)^2 + \left(\frac{\delta Y_{\text{SIMC}}}{Y_{\text{SIMC}}}\right)^2} = 0.663.$$

Thus the toy result is $R = 1.30 \pm 0.66$ (stat). A real analysis must also propagate background-estimate uncertainty and correlated systematic uncertainties; $\sqrt{\text{sum}(f\text{Weight}^2)}$ alone is only the finite-SIMC-sample component.

9. Data corrections versus effects modeled by SIMC

A practical separation is:

Usually corrected or subtracted on data	Usually represented in SIMC/configuration
accumulated beam charge and charge calibration	charge-to-luminosity conversion
DAQ and electronic live time	generated phase-space volume
trigger efficiency	reaction cross-section model
tracking and detector efficiencies	spectrometer optics and apertures
PID efficiency and misidentification	multiple scattering and energy loss
random coincidences and target-wall backgrounds	internal/external radiation as configured
target boiling, if SIMC uses nominal density	particle decay or survival as configured

This table is a default, not a substitute for reading a specific analysis. For example, if a detector inefficiency is explicitly imposed in a custom SIMC, applying it again to the data/SIMC ratio double counts it. Likewise, one must decide whether target density is corrected in data or changed in the SIMC input.

Radiative and bin-centering corrections require special care. In a direct yield comparison, a radiated SIMC prediction should normally be compared to radiated data; do not also "unradiate" the data unless the analysis is explicitly extracting a Born-level result. Bin centering is often obtained from the model after the yield comparison, rather than applied as an ordinary detector efficiency.

10. Analysis recipe

For every data/SIMC comparison:

1. Define the reported quantity: corrected counts, counts/mC, or another explicitly luminosity-normalized yield.
2. Use matching run settings, target isotope and thickness, spectrometer settings, polarity, reaction channel, and radiation configuration.
3. Apply equivalent cuts to equivalent reconstructed variables. The `recon_hcana` output exists specifically to reduce differences between SIMC-style and `hcana`-style reconstructed physics variables.
4. Subtract data backgrounds and apply only the efficiencies absent from the simulation.
5. For positive-Ngen production SIMC, sum `fWeight` after the analysis cuts:

```
cpp double y_simc = 0.0; double var_simc = 0.0; // for each entry passing the identical
analysis/bin cuts: y_simc += fWeight; var_simc += fWeight * fWeight;
```

6. Match charge conventions: either rescale the SIMC sum to the data charge, or divide both results by their respective charges.
7. Compare both absolute normalization and distribution shapes.
8. Quote finite-data, finite-SIMC, background, efficiency, luminosity, acceptance/model, and other relevant uncertainties separately before forming the total.

11. Sanity checks

- Charge scaling: doubling EXPER%charge doubles luminosity, normfac, every fWeight, and the absolute SIMC yield. Dividing by charge removes that factor.
- Monte Carlo convergence: increasing positive Ngen ordinarily increases both Ngen and Ntried. The factor per event shrinks while more events are summed, leaving the expectation value unchanged and reducing statistical fluctuations.
- Acceptance: lower simulated acceptance increases Ntried/Ngen; failed throws contribute zero but remain represented by the Ntried denominator.
- Raw Weight: sum(Weight) lacks luminosity, generated volume, and the attempted-throw normalization, so it is not a predicted count.
- Selected yield: sum(fWeight) after a bin's cuts is the positive-Ngen SIMC prediction for that bin.
- Cut equivalence: matching cut names is insufficient if data and SIMC variables were reconstructed differently; compare definitions and units.
- Special modes: negative ngen and doing_phsp do not automatically obey the production fWeight workflow described here.

12. Compact answer to the six motivating questions

1. Why compare yields? Because both are estimates of the same accepted event-rate integral. Data measure it; SIMC integrates a model of it.
2. What is a SIMC yield? The sum of normalized event contributions. A

charge of 1 mC is a convenient per-mC convention, not a simulated charge deposit and not an immutable SIMC assumption.

3. What are Ntried and Ngen? Ntried counts all random attempts.

Positive Ngen requests that many successful events; negative Ngen requests that many attempts.

4. What is normfac? In normal positive-Ngen production,
 $\text{luminosity} * (\text{Ngen}/\text{Ntried}) * \text{genvol}$.

5. What is Weight? The event's differential physics integrand,
 including the configured cross section, Jacobian, generation correction, and applicable nuclear/target/survival factors.

6. Why $\text{Weight} * \text{normfac} / \text{Ngen}$? The Ngen cancels the success count inside normfac, leaving $\text{Weight} * \text{luminosity} * \text{genvol} / \text{Ntried}$, the standard Monte Carlo contribution. Summing that quantity—not taking one event's value—gives the SIMC yield.

Source map for this repository

- simc.f: target/luminosity calculation, attempted-event loop, Ngen behavior, generation volume, and normfac.
- event.f: generated variables, event restrictions, Jacobians, cross-section completion, and construction of main%weight.
- structures.inc: definitions of weight, gen_weight, jacobian, charge, ngen, and ntried-related state.
- infiles/*.inp: user-facing ngen, EXPER%charge, target, kinematic, and acceptance inputs.
- util/recon_hcana/recon_hcana.C: construction of $\text{fWeight} = \text{Weight} * \text{normfac} / \text{Ngen}$ for positive Ngen.